

3 Within a population, the variation for one characteristic is usually the result of genetic and environmental causes.

(a) Meiosis is one source of genetic variation.

(i) Describe the events that take place during prophase I of meiosis in an animal cell.

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(ii) Explain how independent assortment of homologous chromosomes leads to genetic variation during meiosis I.

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(b) Both genetic and environmental factors contribute to variation in body mass in a bird population.

One cause of variation in body mass in chickens is infection with the gut protoctist *Eimeria*. When fed with the same diet, infected chickens have a lower gain in body mass than healthy chickens in the same population.

(i) Suggest reasons why infected chickens have a lower gain in body mass than healthy chickens.

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(ii) Suggest **one** environmental factor that affects the growth and reproduction of *Eimeria*.

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.....[1]

(c) A study was carried out to investigate the effect of treating infected chickens with extracts from a plant, *Bidens pilosa*.

B. pilosa is used in traditional medicine for the treatment of some infectious diseases.

- Three populations, each containing 25 chickens infected with *Eimeria* were observed.
- The body mass of each chicken was measured at the start of the study.
- Each population was given a different treatment for 56 days:
 - standard diet with no *B. pilosa* extract
 - standard diet with low dose of *B. pilosa* extract
 - standard diet with high dose of *B. pilosa* extract.
- The body mass of each chicken was measured at the end of the study and the gain in body mass over the 56 days was calculated.
- The mean gain in body mass was calculated for each population.

The results are shown in Table 3.1.

Table 3.1

treatment	mean gain in body mass / g
no <i>B. pilosa</i>	1773.0±23.9
low dose of <i>B. pilosa</i>	2093.1 ±34.2
high dose of <i>B. pilosa</i>	2033.3 ±29.9

A statistical analysis of the results of the study of these three populations confirmed that there was a significant difference in mean gain in body mass between low dose *B. pilosa* and high dose *B. Pilosa* extract treatments.

(i) Name a statistical test that could be used to analyse the results of this study.

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- (ii) Sketch on Fig. 3.1 curves to show the pattern of variation for gain in body mass in chickens treated with low dose *B. pilosa* extract and in chickens treated with high dose *B. pilosa* extract.

Label one curve '**low dose**' and the other curve '**high dose**'.

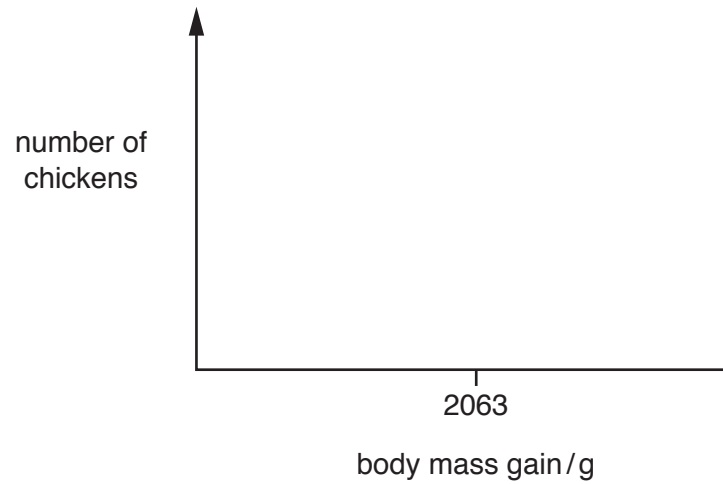


Fig. 3.1

[3]

[Total: 13]